VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

| Reg. Number: | 23BCE1674 |
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# Continuous Assessment Test (CAT) – I AUGUST 2025

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|---|---|---|---|---|---|
| Programme | : | B. Tech Computer Sci Core | Semester | : | Fall 25-26 |
| Course Code & Course Title | : | BCSE301L & Software Engineering | Class Number | : | CH2025260102452<br>CH2025260100690<br>CH2025260100688<br>CH2025260102449<br>CH2025260100691<br>CH2025260100689<br>CH2025260102454 |
| Faculty | : | Dr.SATHYARAJASEKARAN K<br>Dr.ALOK CHAUHAN<br>Dr.JAYANTHI R<br>Dr.PARKAVI K<br>Dr.LAKSHMI HARIKA PALIVELA<br>Dr.BERIN SHALU S<br>Dr.ELAKYA R | Slot | : | E1+TE1 |
| Duration | : | 90 Minutes | Max. Mark | | 50 |

**General Instructions:** < Use this space to provide additional information such as graph sheet, data book etc.>

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

**Answer all questions**

| Q. No | Sub Sec. | Description | Marks | CO | BT Level |
|---|---|---|---|---|---|
| 1 | | The Public Works Department (PWD) aims to create a unified portal that will enable citizens across Tamil Nadu to file complaints related to their respective districts. The system will authenticate users using Aadhaar card details before complaint registration. While experts recommend utilizing the Waterfall Model of Software Development Life Cycle (SDLC) for its structured approach, you, as the Project Manager, prefer the iterative and flexible approach provided by the Agile Model. | 10 | CO1 | Apply |
| | a) | Conduct a detailed analysis and justify your decision to adopt the Agile Model over the Waterfall Model for the PWD project. (5 Marks) | | | |
| | b) | Compare and contrast both models in terms of their adaptability to changing requirements, risk management, and iterative delivery, considering the scope, complexity, and potential challenges in the development of the system. (5 Marks) | | | |
| 2 | a) | TechTrend Innovations is developing a CRM system for a retail client | | | Analyze |

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| | | with a 3-month initial release deadline, requiring frequent updates, high-quality code, and continuous delivery. The team has faced miscommunication, delayed feedback, and rework in past projects. Discuss how Extreme Programming (XP) practices can be applied to this project to address these challenges. In your answer, explain at XP phases and analyze their role in improving communication, reducing rework, and ensuring timely delivery with a neat diagram. (6 Marks) | 10 | CO1 | |
| | b) | List and briefly explain two Scrum activities that could be applied in TechTrend Innovations. (4 Marks) | | | |
| 3 | a) | John is the project documents controller for a Telecom Network Deployment (TND) project that involves developing software for network management. He is tasked with creating a Work Breakdown Structure (WBS) for the software development phase of the project that includes the following major deliverables: Requirements Gathering, Software Design, Coding, Testing, Deployment, and Stakeholder Communication. Break each delivery down into specific tasks and subtasks to ensure a clear and manageable structure for the TND project. (6 Marks) | 10 | CO1 | Analyze |
| | b) | Identify at least four major milestones for the software development phase of a Telecom Network Deployment Project to ensure the project remains on track. (4 Marks) | | | |
| 4 | | Calculate the function point, productivity, documentation, and cost per function for a Hospital Management Software System (HMSS) with multiple Processing Factors: 3, 2, 5, 1, 0, 4, 2, 3, 4, 5, 2, 1, 3. | 10 | CO2 | Evaluate |

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|---|---|
| Number of EI (Avg) | 18 |
| Number of EO (High) | 08 |
| Number of EQ (Low) | 12 |
| Number of ILF (High) | 04 |
| Number of EIF (Avg) | 03 |
| Effort | 42 person-months |
| Software technical documents | 280 pages |
| User-related documents | 150 pages |
| Budgeting/Cost | $8,100 per month |

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| | | From the above given data,<br>Calculate the Unadjusted Function Points (UFP). (2 marks) | | | |
| | a) | Calculate Value Adjustment Factor (VAF). (2 marks) | | | |
| | b) | Calculate Adjusted Function Points (AFP). (2 marks) | | | |
| | c) | Calculate Productivity, Documentation Size per AFP, and Cost per | | | |
| | d) | FP. (4 marks) | | | |
| 5 | | An Online Food Delivery System mobile app is being developed to allow users to browse menus, place orders, make secure payments, and track deliveries in real-time. The app includes distinct user profiles for customers, delivery personnel, restaurant managers, and system administrators. It supports personalized recommendations, scheduled orders, and handles sensitive data like payment details and personal addresses. Ensuring security and role-based access control is critical. | 10 | CO3 | Analyze |

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| | a) | Identify the Functional and Non-Functional Requirements for the above system and represent them using a suitable Software Requirement Specification (SRS) template. (5 Marks) | | | |
| | b) | Explain how you would engage different stakeholders (e.g., customers, delivery staff, restaurant owners, developers, payment gateway providers) in the Requirements Engineering process for the Online Food Delivery System mobile app. (5 Marks) | | | |

**********All the best **********

Six Sigma ISO

CMMI

CMM

func and non functin

SCUM

test plan

web based testing

regression testin

Cohesion and coupling

mutation testing

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# Continuous Assessment Test (CAT) – I AUGUST 2025

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|---|---|---|---|---|---|
| Programme | : | B.Tech Computer Science and Engineering | Semester | : | Fall 2025-26 |
| Course Code & Course Title | : | BCSE301L – Software Engineering | Class Number | : | CH2025260100696<br>CH2025260100697<br>CH2025260100693<br>CH2025260102455<br>CH2025260102456<br>CH2025260102457<br>CH2025260100694 |
| Faculty | : | Dr. ALOK CHAUHAN<br>Dr. LAKSHMI HARIKA PALIVELA<br>Dr. JAYANTHI R<br>Dr. K PARKAVI<br>Dr. R ELAKYA<br>Dr. KUMARAN K<br>Dr. N GANESH | Slot | : | E2 + TE2 |
| Duration | : | 1 Hour 30 Minutes | Max. Mark | | 50 |

**General Instructions:**

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

**Answer all questions**

| Q. No | Sub Sec. | Description | Marks | CO | BT Level |
|---|---|---|---|---|---|
| 1 | | The system helps the Tamilnadu police officers to keep track of the work such as to investigate particular cases, to patrol particular areas and to attend court cases. Some work assignments are regular ongoing assignments, while others are time bound. The system information is updated by the logistics administrator, but individual officers have an interface to display their assigned work. As a project manager, you are aware of the following facts:<br>i) Initial software requirements of above system are reasonably well defined.<br>ii) Availability of limited staff resource and to adhere to the business deadline<br>iii) It is for short duration risky project. | | CO1 | Analyze |
| | A | Identify a suitable process model for the given scenario and justify your choice. | 3 | | |
| | B | Illustrate the architectural diagram with suitable explanations for the chosen model. | 7 | | |
| 2 | A | "A client is the most superior part of an organizational unit". Justify the statement. | 2 | CO1 | Understand |

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|---|---|---|---|---|---|
| | B | Demarcate the roles of customer, end user and the stakeholders in an agile project and elucidate the varied communication languages that the analyst speaks with the client and the development team. | 8 | | |
| 3 | | Assume that you are a software project manager for a project that involves development of a new Customer Relationship Management (CRM) system. The project involves use cases to cover various functionalities such as customer data management, sales reporting, and communication tools. Consider that your team is currently in the estimation phase and has identified the following use cases for the CRM system such as adding new customer, Updating Customer information, Tracking Sales Leads, Generating Sales Report and Sending communication Emails. Assume Simple Use case will take 40 Hours, Moderate use Case will take 80 hours and the Complex Use case takes 160 hours. | | CO2 | Analyze |
| | A | With the details stated above, classify each of the use cases in the CRM system as simple, moderate, or complex, and provide an estimate of the effort required for each use case. | 8 | | |
| | B | Elucidate the rationale behind choosing the use cases. | 2 | | |
| 4 | | The engineering team often receives unpredictable change requests from both marketing and customer service teams. These requests range from urgent bug fixes in payment processing to adding small but high visibility features before campaigns. The company wants to improve the predictability of delivery timelines without losing flexibility in handling unplanned work. | | CO2 | Understand |
| | A | Identify and discuss in detail the combination of agile practices that would be the best fit for this project. | 6 | | |
| | B | Mention the Scrum ceremonies followed in a particular Sprint. | 4 | | |
| 5 | | A mobile application is being developed for a Smart Healthcare Management System that will allow patients to book appointments, consult doctors through video calls, order medicines and access electronic medical records. The system will also have modules for doctors, pharmacists, lab technicians and administrators. | | CO3 | Apply |
| | A | Identify at least four functional requirements and four non-functional requirements for the system. | 4 | | |
| | B | Select three requirement elicitation techniques for this system, justify their selection, and explain how they will be applied for different stakeholders. | 6 | | |

**********All the best **********